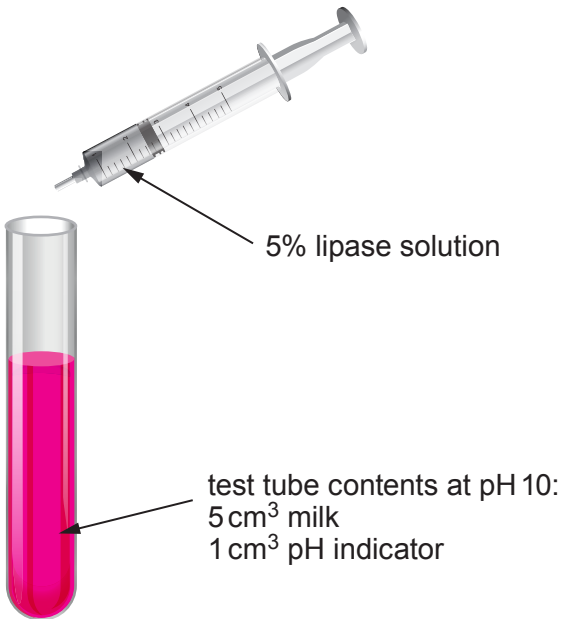


- (b) A class of students investigated how lipase activity changes with temperature. They used a pH indicator that is pink in alkaline solutions of about pH 10. When the pH drops below pH 8 it goes colourless. A solution of full-fat milk, lipase and indicator at pH 10 will change from pink to colourless as the **fat** in milk is broken down producing **fatty acids**. This reduces the pH to below 8. The time taken for this reaction to occur is affected by temperature.

The students worked in pairs and set up the apparatus as shown in **Image 6.2**.

Image 6.2



Each pair of students investigated a different temperature.

- The test tube and syringe were placed in a water bath.
- At 5 minutes, 1 cm³ of lipase solution from the syringe was added to the test tube.
- The time taken for the solution in the test tube to change from pink to colourless was recorded.
- The experiment was carried out three times for each temperature.

The class results are shown in **Table 6.3**.

Table 6.3.

Temperature (°C)	Time for indicator to become colourless (s)				Rate of reaction 1 ÷ mean time (per second)
	Trial 1	Trial 2	Trial 3	Mean	
0 (ice bath)	no change	no change	no change	no change	0.00
20	8.0	6.0	7.0	7.00	0.14
40	5.0	4.0	5.0	4.67	0.21
60	10.0	9.0	9.0	9.33	
80	no change	no change	no change	no change	0.00



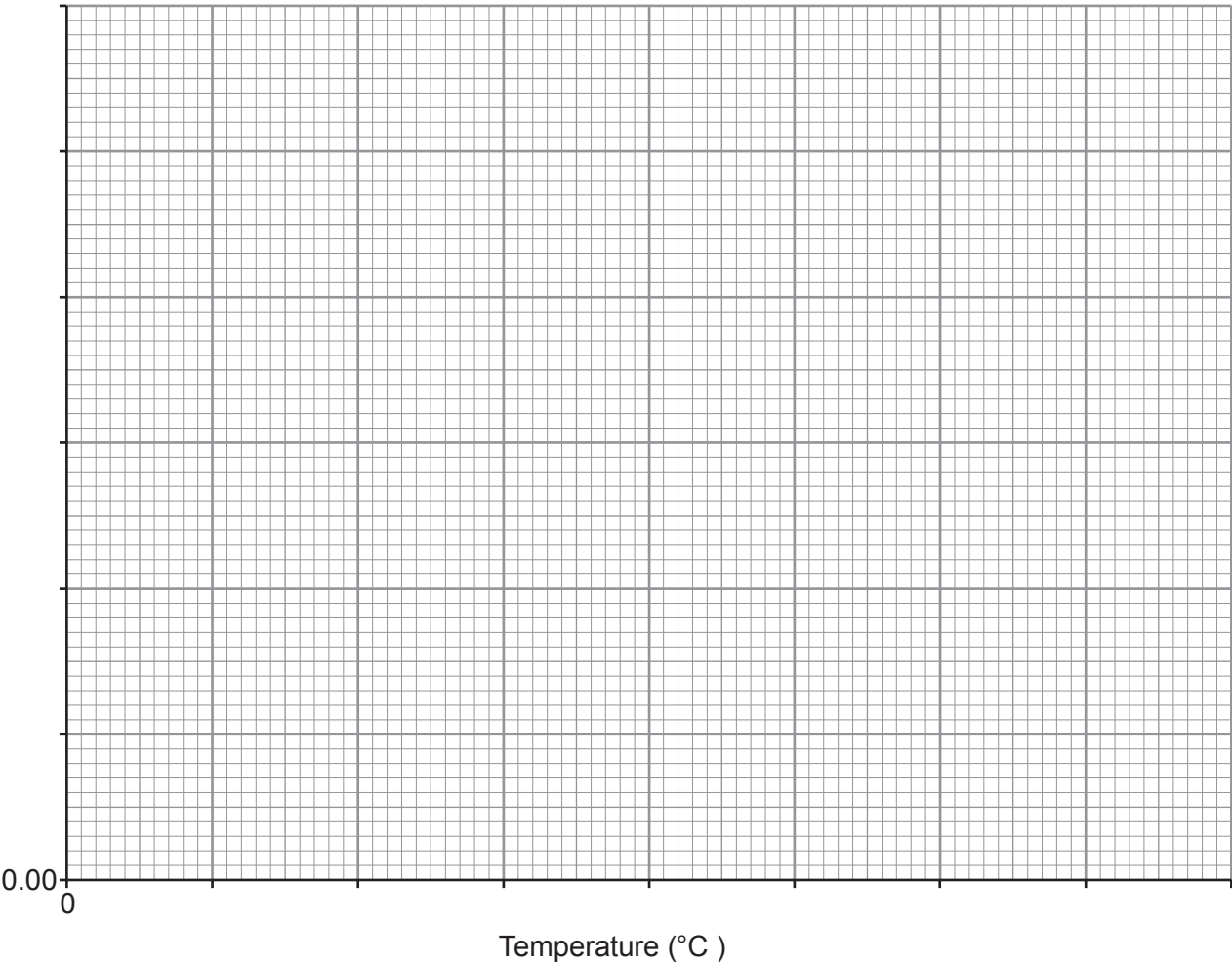
Examiner
only

- (i) The mean time for each temperature can be converted to rate of reaction by calculating $1 \div \text{mean time}$.
Complete Table 6.3 by calculating the rate of reaction at 60 °C. **Give your answer to 2 decimal places.**
Space for working.

[1]

- (ii) On the grid below, draw a **line graph** of **rate of reaction** against temperature. You should join the plots with a ruler.

[4]



- (iii) Describe the effect of temperature on the rate of reaction of lipase.

[1]

.....

.....



Examiner
only

(iv) Explain the effect of temperature on the rate of reaction between 0°C and 40°C. [2]

.....

.....

.....

.....

(c) State why the test tube and syringe were left in the water bath for 5 minutes before the lipase solution was added to the test tube. [1]

.....

.....

(d) Suggest why skimmed milk could not be used in this experiment. [1]

.....

.....

12

